Therefore w.h.p., $|S_j|$ will grow at a rate $(r-1.1)$ once it reaches a size exceeding 100. We must therefore estimate the number of times that this size is not reached. We can bound this as follows. If S_j never reaches 100 in size then some time in the construction of the first $\log_{r-1} 100$ S_j's there will be an edge discovered between an S_j and an excluded vertex. The probability of this can be bounded by $100 \cdot O(n^{-3/11}) = O(n^{-3/11})$. So, if β denotes the number of i that fail to produce S_t of size $n^{3/5}$ then

$$\Pr(\beta \geq 20) \leq o(n^{-3}) + \binom{n^{1/10} \log n}{20} \cdot O(n^{-3/11})^{20} = o(n^{-3}).$$

Thus w.h.p. there will be at least $n^{1/20} - 20 > n^{1/21}$ of the u_i from which we can grow a tree with $n^{3/5}$ leaves $L_{i,y}$ such that all these trees are vertex disjoint from each other and R.

By the same argument we can find at least $n^{1/21}$ of the v_i from which we can grow a tree $L_{i,x}$ with $n^{3/5}$ leaves such that all these trees are vertex disjoint from each other and R *and the trees grown from the* u_i. We then observe that if $e(L_{i,x}, L_{i,y})$ denotes the edges from $L_{i,x}$ to $L_{i,y}$ then

$$\Pr(\exists i : e(L_{i,x}, L_{i,y}) = \emptyset) \leq n^{1/20} \left(1 - \frac{(r-1)n^{3/5}}{rn/2}\right)^{(r-1)n^{3/5}} = o(n^{-3}).$$

We can therefore w.h.p. choose an edge $f_i \in e(L_{i,x}, L_{i,y})$ for $1 \leq i \leq n^{1/21}$. Each edge f_i defines a path Q_i from x to y of length at most $2\log_{r-1} n$. Let Q_i' denote that part of Q_i that goes from $u_i \in A_{t_0}$ to $v_i \in B_{t_0}$. The path Q_i will be rainbow colored if the edges of Q_i' are rainbow colored and distinct from the colors in the path from x to u_i in T_{x,t_0} and the colors in the path from y to v_i in T_{y,t_0}. The probability that Q_i' satisfies this condition is at least $\left(1 - \frac{2\log_{r-1} n}{q}\right)^{2\log_{r-1} n}$. Here we have used (13). In fact, using (13) we see that

$$\begin{aligned}\Pr(\nexists i : Q_i \text{ is rainbow colored}) &\leq \left(1 - \left(1 - \frac{2\log_{r-1} n}{q}\right)^{2\log_{r-1} n}\right)^{n^{1/21}} \\ &\leq \left(1 - \frac{1}{n^{4/(rK_1^2)}}\right)^{n^{1/21}} = o(n^{-3}).\end{aligned}$$

This completes the case where x, y are both tree-like and $T_x \cap T_y = \emptyset$.

2.6.2 Tree-like but not disjoint

Suppose now that x, y are both tree-like and $T_x \cap T_y \neq \emptyset$. If $x \in T_y$ or $y \in T_x$ then there is nothing more to do as each root to leaf path of T_x or T_y is rainbow.

Let $a \in T_y \cap T_x$ be such that its parent in T_x is not in T_y. Then a must be a leaf of T_y. We now bound the number of leaves λ_a in T_y that are descendants of a in T_x. For this we need the distance of y from T_x. Suppose that this is h. Then

$$\lambda_a = 1+(r-2)+(r-1)(r-2)+(r-1)^2(r-2)+\cdots+(r-1)^{k_r-h-1}(r-2) = (r-1)^{k_r-h}+1.$$

Now from Lemma 5 we see that there will be at most two choices for a. Otherwise, $T_x \cup T_y$ will contain at least two cycles of length less than $2k_r$. It follows that w.h.p. there at most $\lambda_0 = 2((r-1)^{k_r-h}+1)$ leaves of T_y that are in T_x. If $(r-1)^h \geq 201$ then $\lambda_0 \leq |S_y|/10$. Similarly, if $(r-1)^h \geq 201$ then at most $|S_x|/10$ leaves of T_x will be in T_y. In which case we can use the proof for $T_x \cap T_y = \emptyset$ with S_x, S_y cut down by a factor of at most $4/5$.

If $(r-1)^h \leq 200$, implying that $h \leq 5$ then we proceed as follows: We just replace k_r by $k_r + 5$ in our definition of T_x, T_y, for these pairs. Nothing much will change. We will need to make q bigger by a constant factor, but now we will have $y \in T_x$ and we are done.

2.6.3 Non tree-like

We can assume that if x is non tree-like then T_x contains exactly one cycle C. We first consider the case where C contains an edge e that is more than distance 5 away from x. Let $e = (u, v)$ where u is the parent of v and u is at distance 5 from x. Let $\widehat{T}_x$ be obtained from T_x by deleting the edge e and adding two trees H_u, H_v, one rooted at u and one rooted at v so that $\widehat{T}_x$ is a complete $(r-1)$-ary tree of height k_r. Now color H_u, H_v so that Lemma 3 can be applied. We create $\widehat{T}_y$ from T_y in the same way, if necessary. We obtain at least $(r-1)^{2k_r}/5$ pairs. But now we must subtract pairs that correspond to leaves of H_u, H_v. By construction there are at most $4(r-1)^{2k_r-5} \leq (r-1)^{2k_r}/10$. So, at least $(r-1)^{2k_r}/10$ pairs can be used to complete the rest of the proof as before.

We finally deal with those T_x containing a cycle of length 10 or less, no edge of which is further than distance 10 from x. Now the expected number of vertices on cycles of length $k \leq 10$ is given by

$$k\binom{n}{k}\frac{(k-1)!}{2}\binom{r}{2}^k 2^k \frac{\Psi(rn-2k)}{\Psi(rn)} \sim \frac{(r-1)^k}{2k},$$

where $\Psi(m) = m!/(2^{m/2}(m/2)!)$.

It follows that the expected number of edges μ that are within 10 or less from a cycle of length 10 or less is bounded by a constant. Hence $\mu = o(\log n)$ w.h.p. and

we can give each of these edges a distinct new color after the first round of coloring. Any rainbow colored set of edges will remain rainbow colored after this change.

Then to find a rainbow path beginning at x we first take a rainbow path to some x' that is distance 10 from x and then seek a rainbow path from x'. The path from x to x' will not cause a problem as the edges on this path are unique to it.

3 The case $d = 3$

An easy generalization of the example in Figure 1 shows that Lemma 2 does not extend to binary trees. It indicates that $\kappa_{\ell,2} \leq 2^\ell$ and not $\Omega(2^{2\ell})$ as we would like. In this case we have not been able to prove Corollary 4. Note that while the example

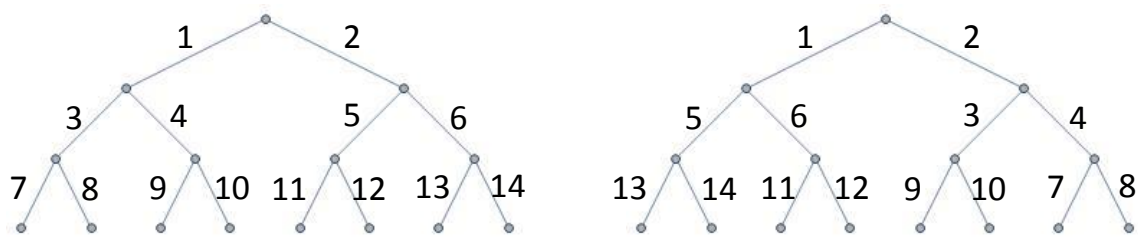

Figure 1: Two rainbow trees T_1 and T_2 with $m(T_1, T_2) = 2^3$.

shows that $m(T_1, T_2) = 2^\ell$, it does show there is a bijection f between the leaves of the two trees so that $P_{x,T_1} \cup P_{f(x),T_2}$ is rainbow. In fact, an elegant probabilistic argument due to Noga Alon [1] shows that with the hypothesis of Lemma 2, there are always sets $S_i \subseteq L_i$ and a bijection $f : S_1 \to S_2$ such that (i) $|S_1| = |S_2| = \Omega(2^\ell)$ and such that (ii) $x \in S_1$ implies that $P_{x,T_1} \cup P_{f(x),T_2}$ is rainbow. This is a step in the right direction and it can be used to show that $O\left(\left(\frac{\log n}{\log\log n}\right)^2\right)$ colors suffice, beating the bound implied by [2].

4 Conclusion

We have shown that w.h.p. $rc(G(n,r)) = O(\log n)$ for $r \geq 4$ and $r = O(1)$. Determining the hidden constant seems challenging. We have seen that the argument for $d \geq 4$ cannot be extended to $d = 3$ and so this case represents a challenge.

At a more technical level, we should also consider the case where $r \to \infty$ with n.